Spatiotemporal Dual Priors Guided Diffusion for Remote Sensing Spatiotemporal Fusion

Baoluo Zhu, Huihui Song, Kaihua Zhang, and Guangcan Liu

***Abstract*—Remote sensing spatiotemporal fusion (STF) aims to integrate remote sensing images with complementary temporal and spatial resolutions to enhance image quality with high spatiotemporal resolutions. Conditional diffusion models have shown promising performance in STF by learning the data probability distribution to achieve accurate fusion results. However, existing conditional diffusion models for STF directly use the limited image exemplars as conditional signals, limiting their ability to extract coherent features across images with extended temporal gaps and significant resolution differences. To address this challenge, this article presents a spatiotemporal dual priors guided diffusion (DPDiff) framework for STF. The DPDiff has three key modules including a prior extraction network (PENet), a residual diffusion framework (RDF), and a difference-aware module (DAM). First, a lightweight convolutional neural network (CNN)-based PENet is designed to learn the spatiotemporal dual priors as conditional signals to control the diffusion process. The dual priors include a spatial mapping prior (SMP) between the coarse and fine images at the same time, and a temporal change prior (TCP) between the coarse and fine images at different times. The SMP and the TCP are reciprocal to ensure precise control of the diffusion process. Then, an RDF is adapted to learn distribution differences between high temporal resolution and high spatial resolution images, rather than directly reconstructing the entire image. Third, a DAM is introduced to align the CNN-predicted space with the ground-truth (GT) space. The experimental results on four benchmark datasets—CIA, LGC, WaterBank, and the global E-SMILE—demonstrate that DPDiff achieves superior generation fidelity and quantitative performance across diverse regions and platforms, outperforming state-of-the-art methods. The source code is available at https://github.com/paul623/DPDiff**

***Index Terms*—Conditional diffusion, deep learning, remote sensing, spatial–temporal fusion.**

I. Introduction

REMOTE sensing imagery is vital for applications in agriculture [1], [2], forestry [3], and ecology [4], [5] due to its ability to capture detailed surface information and provide extensive temporal coverage [6]. However, achieving both high spatial and high temporal resolutions with a single optical sensor is challenging because of cost, technical constraints, and persistent cloud cover that often causes missing data [7]. To overcome these issues, remote sensing spatiotemporal fusion (STF) has been developed. STF integrates high spatial resolution images (i.e., fine images), high temporal resolution images (i.e., coarse images), and the coarse image at the target time to generate fine images for that time, thereby enhancing data continuity and resolution [8].

Currently, the STF field confronts two primary challenges: temporal uncertainty and spatial uncertainty [12]. Temporal uncertainty refers to the unpredictability of the time gap between the target and reference time. This can impede models from extracting accurate features for target image reconstruction, especially in scenarios with significant phenological or land-cover changes, where feature transfer becomes more complex. Spatial uncertainty refers to the significant resolution differences in remote sensing images (e.g., Landsat versus MODIS satellites, 16×), which are much greater than those typically encountered in image super-resolution tasks (typically 2×–8×), making it challenging for models to capture such complex mapping relationships accurately [13].

To tackle these challenges, the current research lines can be broadly categorized into convolutional neural network (CNN)/Transformer-based methods [13], [14], [15], [16], [17], [18], [19] and generative model-based methods [12], [20], [21], [22], [23], [24]. The former primarily focuses on modeling the mapping relationship between coarse and fine images, utilizing convolutional neural networks [25] or Transformers [26] to extract features at different scales for fusion. These methods are straightforward to implement and perform well in areas with gradual land-cover changes, but their performance deteriorates in the presence of temporal uncertainty, particularly with rapid phenological or land-cover changes. In contrast, the latter takes the representative generative adversarial networks (GANs) [27] and the diffusion model [28] as representative, which focus on learning the distribution of images and leverages powerful generative capabilities to generate target images from the probability distribution, rather than directly learning an exact mapping from coarse to fine, thereby alleviating issues caused by temporal uncertainty. Among the generative-based methods [12], [20], [21], [22], [23], [24], the conditional diffusion model [28] has demonstrated better performance than GAN [27] and a more stable training process in super-resolution and feature reconstruction tasks, which has gradually been applied to STF [29]. However, achieving precise control

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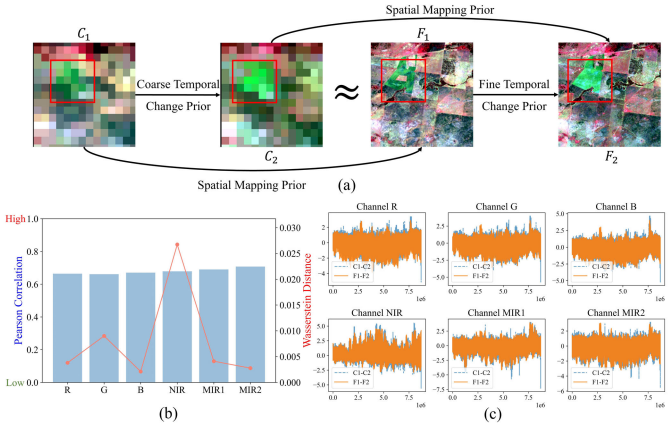


Fig. 1. Illustration of the spatiotemporal dual priors composed of a TCP and an SMP. (a) TCP assumes that the temporal changes in coarse images approximate those in fine images, ensuring feature temporal consistency, while the SMP guides coarse-to-fine transition for spatial consistency. (b) Validation of the SMP on LGC dataset [9]: the Pearson correlation coefficient [10] and the Wasserstein distance [11] reveal a strong association between coarse and fine mappings, supporting target mapping prediction. (c) Validation of the TCP on the LGC dataset: normalized differences confirm that coarse-image change trends closely to those in fine images.

over the generation process remains a significant challenge in conditional diffusion models, particularly in extracting coherent features for fine-grained control. Existing methods typically use limited image exemplars as direct conditional inputs to implicitly model the relationship between coarse and fine images, which constrains the accurate reconstruction of fine details across extended temporal gaps and significant resolution differences, ultimately degrading the quality of the generated results [22], [30]. Moreover, the existing diffusion-based STF models [12], [22], [23], [24] still adhere to the traditional diffusion process, initiating denoising from pure Gaussian noise, which is unnecessary in practical applications [30], [31], [32]. Since the coarse image at the target time already contains certain low-frequency information, effectively integrating the coarse image into the denoising process can enhance sampling efficiency while ensuring the accuracy of the generated image.

Thus, to mitigate spatiotemporal uncertainties, it is essential to establish reliable priors that capture the correlations between known and target images. This article introduces spatiotemporal dual priors, including the temporal change prior (TCP) and the spatial mapping prior (SMP), to enhance the diffusion model in spatial reconstruction and temporal change modeling, as illustrated in Fig. 1(a). These priors are derived from statistical observations, with the LGC dataset [9] serving as an example. First, we establish the SMP by examining the relationship between coarse and fine images captured simultaneously by two satellites, MODIS [33] and Landsat-8 [34]. To quantify this relationship, we compute the differences between coarse and fine image pairs across different times and assess their feature similarity and distributional alignment using the Pearson correlation coefficient [10] and the Wasserstein distance [11], respectively. As shown in Fig. 1(b), the correlation coefficients across all spectral channels exceed 0.5, indicating strong feature associations, while the Wasserstein distances remain below 0.03, indicating similar distributional characteristics. These results confirm that SMP effectively constrains the spatial correspondences between known and target images. Next, we investigate whether coarse–fine image pairs from different times exhibit the same change trends, referred to as TCP. Specifically, we analyze temporal differences in coarse images and compare them with the corresponding differences in fine images across different times. As depicted in Fig. 1(c), the observed change trends in coarse images align closely with those in fine images across all spectral channels. This finding suggests that TCP extracted from coarse images can effectively capture the temporal dynamics of fine images, thereby providing essential temporal guidance for the diffusion process. Consistent patterns are also observed on other satellite platforms such as Sentinel-2 and Sentinel-3, suggesting the general applicability of the proposed priors.

Given the demonstrated effectiveness of the proposed SMP and TCP, we develop a spatiotemporal dual priors guided diffusion (DPDiff) framework for STF. A lightweight CNN-based prior extraction network (PENet) is first designed to derive these priors, which are subsequently injected into the diffusion process for conditional guidance. To further improve sampling efficiency, we adopt the residual diffusion strategy [35], allowing the model to focus on learning the residual distribution between coarse and fine images rather than reconstructing the entire image from pure Gaussian noise. Additionally, a difference-aware module (DAM) is introduced to adaptively correct the discrepancies between the prior-predicted and target spaces, thereby enhancing detail preservation and temporal consistency. Distinct from STFDiff [12] and resnoise diffusion [30], which mainly emphasize diffusion efficiency or spatial fidelity, DPDiff introduces a fundamentally different conditioning paradigm. Instead of merely adopting residual diffusion, DPDiff reformulates it within a dual-prior-guided residual space, where SMP and TCP serve as explicit, learnable constraints that model cross-resolution spatial correlations and temporal evolution trends. This unified design couples prior extraction and residual learning, yielding a diffusion process that is both more efficient and more physically interpretable for STF. Extensive experiments on four benchmark datasets [9], [12], [36] demonstrate that DPDiff consistently outperforms state-of-the-art STF methods in both quantitative metrics and visual quality. Specifically, our contributions are as follows.

1) A lightweight CNN-based PENet is designed to learn SMP and TCP as conditional signals. These dual priors guide the diffusion process to handle images with extended temporal gaps and significant resolution differences.
2) A residual diffusion framework (RDF) is designed that focuses on learning the distribution differences between high temporal resolution and high spatial resolution images. This approach improves sampling efficiency and aligns with the optimization objectives of the PENet.
3) A DAM is developed to adaptively align the CNN-predicted space with the ground-truth (GT) space by learning their intrinsic discrepancies. This module

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compensates for the insufficiency of prior information, enhances the representation of fine details and temporal variations, and improves generation fidelity without excessive dependence on prior features.

The remainder of this article is organized as follows. Section II reviews the related work. Section III provides a detailed introduction to the proposed DPDiff, with an emphasis on the extraction of spatiotemporal priors. Section IV presents the experimental results and analysis. Finally, Section V concludes this article and provides final remarks.

II. Related Work

A. Traditional STF Methods

Weighted function-based methods estimate pixel values in the target fine image by assigning adaptive weights to neighboring pixels in the input images based on their similarity, typically using a weighted averaging strategy. STARFM [37], one of the earliest and most widely used methods in this category, assumes that temporal changes in coarse pixels align with those in fine pixels and that each pixel comprises a single endmember. This assumption enables STARFM to achieve robust predictions in homogeneous landscapes dominated by coarse pixels. Although subsequent advancements [38], [39], [40] have improved performance, their effectiveness in heterogeneous regions remains limited. The primary limitations stem from their oversimplified assumptions about pixel relationships and insufficient consideration of temporal uncertainty [41].

Unmixing-based methods are founded on linear spectral unmixing theory [42] to extract endmember reflectance from reference images and to predict detailed pixel information by decomposing the target coarse pixels. MMT [43] is the first STF method based on unmixing although its spectral decomposition accuracy is relatively limited. Building on this, FSDAF [44] integrates weight function-based approaches with unmixing-based methods and incorporates spatial interpolation techniques, significantly enhancing performance in heterogeneous regions or areas experiencing rapid land-cover changes. Nevertheless, in practical applications, the assumptions underlying unmixing often diverge from the complex mixing processes of endmembers in real-world scenarios, which ultimately results in lower fusion accuracy for most unmixing-based methods [6].

Bayesian-based methods utilize Bayesian estimation theory to construct statistical models for remote sensing images at different times, enabling the generation of fine images. These methods offer greater flexibility, effectively reducing uncertainties introduced by input images and producing highly detailed results. For instance, MDBFM [45] and Ref-BFSM [46] extract change information from coarse images, making them well-suited for handling phenological changes and changes in land-cover types. However, most Bayesian-based methods are designed for specific applications, such as land vegetation detection [47] and normalized difference vegetation index (NDVI) [48]. Furthermore, some methods impose strict requirements on input data [49], limiting their applicability across diverse scenarios.

B. Learning-Based STF Methods

Deep learning-based methods adopt a data-driven approach, allowing automatic prior knowledge extraction and effective modeling of complex nonlinear relationships. Currently, STF methods based on deep learning fall into two categories: CNN/Transformer-based and generative-based methods.

1) CNN/Transformer-Based Methods: STFDCNN [14] adopts a super-resolution paradigm to learn the mapping between coarse and fine images. However, the large resolution gap results in suboptimal reconstruction performance. Subsequently, researchers shift toward fusion-based modeling, focusing on extracting spatial information from fine images and temporal information from coarse images to predict the target image, rather than merely learning the mapping between coarse and fine images. For example, StfNet [15] employs a dual-stream convolutional network to extract temporal change information from coarse image differences while capturing spatial information from fine images, thereby enhancing fusion performance. MSNet [17] replaces the CNN branches in StfNet with a Transformer structure, leveraging the global modeling capabilities of Transformers to further improve the extraction and prediction of temporal correlations. Inspired by traditional methods, SwinSTFM [13] incorporates dynamic adjustments based on land-cover types, optimizing the attention matrix to improve performance in heterogeneous regions. Moreover, ECPW-STFN [18] tackles the STF problem in the frequency domain, using wavelet transforms to learn high- and low-frequency features separately. GSINet [19] is designed to improve spectral fidelity by leveraging spectral characteristics through a combination of intragroup fusion and intergroup interaction. FSDFormer [50] further incorporates a frequency-selected differential mechanism to enhance change representation in dynamic regions.

2) Generative-Based Methods: Although CNN/Transformer-based methods have achieved some success in addressing spatial uncertainty, they are still insufficient in handling issues arising from temporal uncertainty, such as land-cover changes. Consequently, generative model-based approaches have emerged, adopting a "reconstruction" strategy as an alternative to "fusion." GAN-STFM [20] and MLFF-GAN [21] utilize GANs to minimize the resolution gap between coarse and fine images, thereby improving prediction accuracy. DiffSTF [23] introduces diffusion models into the STF for the first time by enhancing the noise prediction network structure. However, due to the lack of explicit conditional constraints, the generated images exhibit deficiencies in fine-grained features, particularly underperforming in metrics like the structural similarity index (SSIM). Similarly, STFDiff [12] improves the noise prediction network and introduces an independent conditional control branch to constrain the generation process. GCM-PDA [51] proposes a generative compensation strategy for progressive difference attenuation, enhancing temporal consistency during multitemporal fusion. However, this implicit control mechanism fails to fully exploit the relationships between reference and target images and does not sufficiently leverage the generative capabilities of

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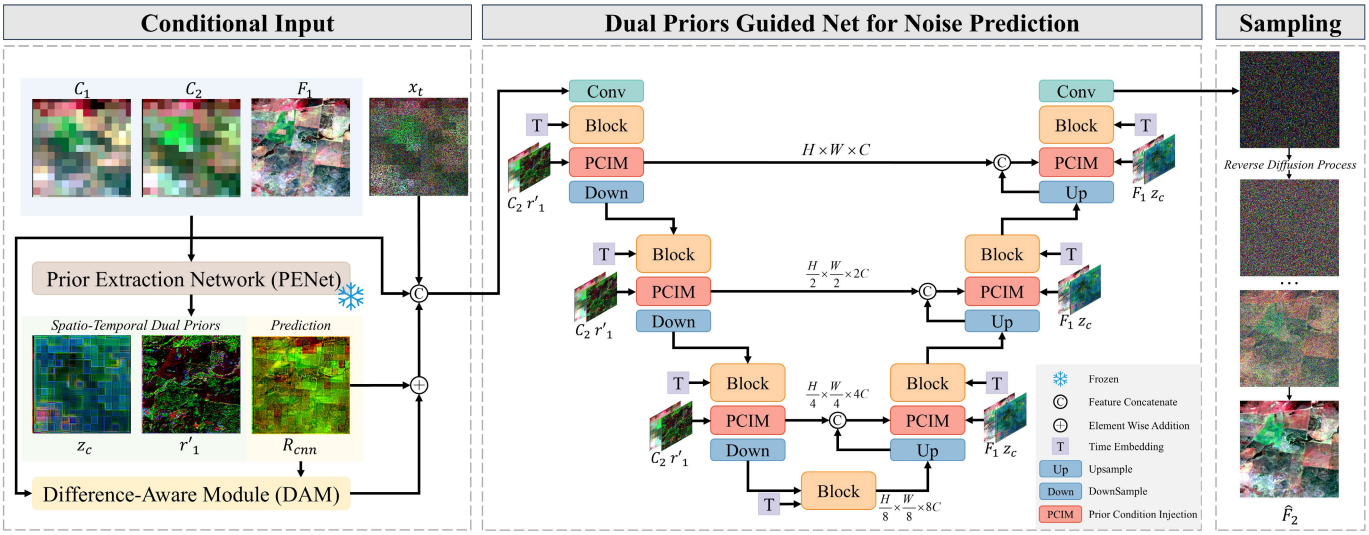


Fig. 2. Pipeline of our proposed DPDiff. It consists of three core steps: conditional input, denoising network, and denoising sampling. In the conditional input step, the known image pairs are passed through a pretrained PENet to extract the SMP $\mathbf{r}'_1$, TCP $\mathbf{z}_c$, and the CNN-predicted $\mathbf{R}_{cnn}$. The DAM then compensates $\mathbf{R}_{cnn}$, and the compensated result, along with the known images, is used as the conditional input to the denoising network. The denoising network employs U-net [52] as the backbone, with PCIM applied in the encoder and decoder to incorporate prior constraints. The denoising process is optimized by a residual design, requiring only five sampling steps to generate the target image.

diffusion models to effectively address temporal uncertainty, leaving room for further improvement.

III. Methods

In this study, we introduce the TCP and the SMP to precisely constrain the diffusion process. This approach not only leverages the powerful generative capability of the diffusion model but also ensures the accuracy of the generated images. As shown in Fig. 2, DPDiff adopts a two-stage network architecture, consisting of three core components: conditional input, a denoising network, and a sampling process. In the first stage, the model takes coarse image $\mathbf{C}_1 \in \mathbb{R}^{H\times W\times C}$ at reference time t_1, fine image $\mathbf{F}_1 \in \mathbb{R}^{H\times W\times C}$ at t_1, and coarse image $\mathbf{C}_2 \in \mathbb{R}^{H\times W\times C}$ at target time t_2 as inputs. It, then, employs PENet to extract the TCP $\mathbf{z}_c \in \mathbb{R}^{H\times W\times 6C}$ and the SMP $\mathbf{r}'_1 \in \mathbb{R}^{H\times W\times 6C}$, generating an intermediate prediction $\mathbf{R}_{cnn} \in \mathbb{R}^{H\times W\times C}$. Here, H is the height, W is the width, and C is the channel. In the second stage, the diffusion model is trained with $\mathbf{C}_1, \mathbf{C}_2, \mathbf{F}_1$, and $\mathbf{R}_{cnn}$, and the priors $\mathbf{z}_c$ and $\mathbf{r}'_1$ as inputs, refining the prediction through the reverse diffusion process to generate the final fine image $\hat{\mathbf{F}}_2$.

A. Preliminary

The conditional diffusion model introduces a conditioning mechanism based on the standard diffusion model [22], enhancing the controllability of the generated results. During the diffusion process, the initial image $x_0 = \mathbf{F}_2$ is gradually corrupted by Gaussian noise, resulting in a noisy image x_t at a given time step t. Subsequently, a noise prediction network is trained to estimate the noise components in x_t, using prior images $\mathbf{F}_1$, $\mathbf{C}_1$, and $\mathbf{C}_2$ as conditional constraints. During the reverse diffusion (denoising) process, the model gradually reconstructs the target image from pure Gaussian noise, guided by the prior images and the trained noise prediction network.

In the STF task, $\mathbf{C}_2$ is essentially a low spatial resolution version of the target image $\mathbf{F}_2$, meaning that $\mathbf{C}_2$ can be regarded as a degraded representation of $\mathbf{F}_2$. Consequently, it already contains certain low-frequency information, making it unnecessary to perform denoising from pure noise as in conventional diffusion methods. To address this, we replace the conventional diffusion process with resnoise diffusion [35], which improves sampling efficiency and enhances the fidelity of the generated image content.

1) Forward Process: First, let the input degraded image $\mathbf{C}_2$ and the GT $\mathbf{F}_2$ be denoted as $\hat{x}_0$ and x_0, respectively. In our framework, we aim for the model to learn the difference between coarse and fine images, i.e., the SMP at time t_2

$$R = x_0 - \hat{x}_0. \tag{1}$$

This design aligns with the optimization objective of PENet, which will be introduced in Section III-B. According to the design of denoising diffusion probabilistic model (DDPM) [28], the forward process can be defined as

$$\begin{aligned} q(x_{1:T} \mid x_0, R) &= \prod_{t=1}^{T} q(x_t \mid x_{t-1}, R) \\ q(x_t \mid x_{t-1}, R) &= N\left(x_t; \sqrt{\alpha_t}x_{t-1} + \left(1-\sqrt{\alpha_t}\right)R, (1-\alpha_t)I\right) \end{aligned} \tag{2}$$

where $\beta_t = 1 - \alpha_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. The noisy image x_t is, then, obtained via

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \left(1-\sqrt{\bar{\alpha}_t}\right)R + \sqrt{1-\bar{\alpha}_t}\varepsilon \tag{3}$$

where $\varepsilon \sim N(0, I)$. To quantify the difference between the predicted x_t and the GT, resnoise diffusion defines a new variable, $res\varepsilon$, and the optimization objective becomes

$$\begin{cases} \text{res}\varepsilon = \varepsilon + \dfrac{\left(1-\sqrt{\alpha_t}\right)\sqrt{1-\bar{\alpha}_t}}{\beta_t}R \\ \mathbb{E}_{x_0,\epsilon,t}\left[\|\text{res}\epsilon - \text{res}\epsilon_\theta(x_t, t)\|^2\right]. \end{cases} \tag{4}$$

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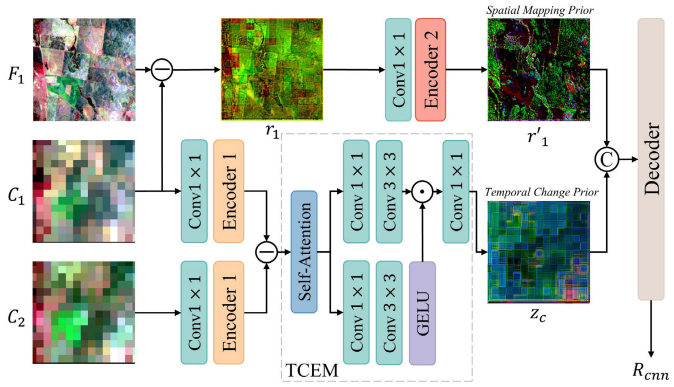


Fig. 3. Architecture of PENet. PENet is first trained independently and then kept frozen during the diffusion training phase to preserve statistically consistent priors. It is designed to extract the coarse-TCP $\mathbf{z}_c$ and the SMP $\mathbf{r}'_1$, which serve as constraints for the diffusion process. The network output $\mathbf{R}_{cnn}$ is optimized to approximate the GT SMP $\mathbf{r}_2$, while the frozen design helps stabilize the subsequent conditional diffusion training.

2) Denoising Process: According to (1) and (3), we obtain $x_t = (2(\bar{\alpha}_t)^{1/2} - 1)x_0 + (1 - (\bar{\alpha}_t)^{1/2})\hat{x}_0 + (1 - \bar{\alpha}_t)^{1/2}\varepsilon$. However, x_0 is unknown during the reverse process. Therefore, resnoise diffusion introduces a smooth equivalence transformation mechanism to eliminate the coefficient of x_0 0, i.e.,

$$\begin{cases} T' = \operatorname{argmin}_{i=1}^{T} \left| \sqrt{\bar{\alpha}_i} - \dfrac{1}{2} \right| \\ x_{T'} = \sqrt{\bar{\alpha}_{T'}}\hat{x}_0 + \sqrt{1 - \bar{\alpha}_{T'}}\varepsilon. \end{cases} \tag{5}$$

For time $t \sim \text{Uniform}(2, \ldots, T')$, we can compute x_{t-1} as

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \left(res\varepsilon_\theta (x_t, \hat{x}_0, t) \right) \right) + \sqrt{\tilde{\beta}_t} z \tag{6}$$

where $z \sim N(0, I)$ and $\tilde{\beta}_t = (1 - \bar{\alpha}_{t-1})/(1 - \bar{\alpha}_t)\beta_t$. Finally, for $t = 1$, we have

$$x_0 = \frac{1}{\sqrt{\alpha_1}} \left(x_1 - \frac{\beta_1}{\sqrt{1 - \bar{\alpha}_1}} res\varepsilon_\theta (x_1, \hat{x}_0, 1) \right). \tag{7}$$

B. Prior Extraction Network

To explicitly constrain the diffusion process, we introduce the TCP and the SMP. The TCP enhances feature consistency by aligning the difference between coarse images at the target and reference times with that of their high-resolution counterparts [53], [54]. Meanwhile, the SMP assumes that the transformation between coarse and fine images remains consistent across different times and regions, thereby regularizing the resolution mapping [37], [55]. Building upon this, we integrate the TCP derived from coarse images at two times and the SMP inferred from known coarse–fine image pairs to further estimate the SMP at the target time [56]. Accordingly, we reformulate the learning objective as predicting the difference between the fine and coarse images at the target time, which effectively captures the spatial mapping relationship at that time. To achieve this, we propose the PENet to extract and incorporate these priors, as illustrated in Fig. 3.

Specifically, in the first step, we extract the coarse TCP (coarse-TCP), denoted as $\mathbf{z}_c$, from the coarse images $\mathbf{C}_1$ and $\mathbf{C}_2$. We assume that there exists a mapping function $f(x)$ that transforms $\mathbf{z}_c$ into the fine TCP (fine-TCP) $\mathbf{z}_f$, such that $\mathbf{z}_f = f(\mathbf{z}_c)$. To achieve this, we utilize a weight-sharing encoder to extract high-dimensional features from $\mathbf{C}_1$ and $\mathbf{C}_2$ and compute their difference. Subsequently, we apply the temporal change enhancement module (TCEM) to further enhance the change features, thereby obtaining $\mathbf{z}_c$. The TCEM consists of two key components: self-attention and a gating mechanism. The self-attention mechanism [26] enables the model to capture long-range dependencies and refine the temporal change features by considering global interactions between pixels. Meanwhile, the gating mechanism adaptively controls the information flow, filtering out irrelevant changes and highlighting meaningful changes. Self-attention is defined as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} \right) \mathbf{V} \tag{8}$$

where $(d_k)^{1/2}$ is a scaling factor, $\mathbf{Q}$ denotes the query, and $\mathbf{K}$ and $\mathbf{V}$ stand for the key and value, respectively. On the other hand, the SMP can essentially be represented as the difference between the fine image and the coarse image. Therefore, for the SMP $\mathbf{r}_1$ at time t_1, we have $\mathbf{r}_1 = \mathbf{F}_1 - \mathbf{C}_1$. To integrate this with the coarse-TCP $\mathbf{z}_c$, we employ a separate encoder to project $\mathbf{r}_1$ into the same feature space as $\mathbf{z}_c$, denoted as $\mathbf{r}'_1$, and then feed both into a decoder to ultimately obtain the predicted $\mathbf{R}_{cnn}$. The optimization objective is to minimize the discrepancy between $\mathbf{R}_{cnn}$ and $\mathbf{r}_2$, i.e., $\mathbf{r}_2 = \mathbf{F}_2 - \mathbf{C}_2$. The overall process can be formulated as

$$\begin{cases} \mathbf{z}_c = \text{TCEM}\left(\text{Encoder}_1(\mathbf{C}_1) - \text{Encoder}_1(\mathbf{C}_2)\right) \\ \mathbf{r}'_1 = \text{Encoder}_2(\mathbf{r}_1) \\ \mathbf{R}_{cnn} = \text{Decoder}\left(\mathbf{z}_c, \mathbf{r}'_1\right). \end{cases} \tag{9}$$

In the design of the loss function, to make PENet focus more on the coarse-to-fine mapping, i.e., the difference between the fine and coarse images, we add a structural loss [57] on top of the pixelwise loss [58]. The two loss functions are described as follows:

$$\begin{aligned} \mathcal{L}_{\text{structure}} &= 1 - \min\left(\text{MS} - \text{SSIM}\left(\mathbf{R}_{cnn}, \mathbf{r}_2\right) + \mu_s, 1\right) \\ \mathcal{L}_{\text{pixel}} &= \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \rho\left(\mathbf{R}(i,j) - r_2(i,j)\right) \end{aligned} \tag{10}$$

where $\rho(x) = (x^2 + \mu_p^2)^{1/2}$, $\mathbf{r}_2 = \mathbf{F}_2 - \mathbf{C}_2$, and $\mathbf{R}_{cnn}$ is the prediction. To stabilize the backpropagation of the loss, we introduce μ_p. Since it is difficult for the structural loss to reach 1, we incorporate μ_s to modulate the effect of samples with very small structural losses during training. The values for μ_p and μ_s are set to 0.001 and 0.05, respectively. Finally, the overall loss function can be expressed as

$$\mathcal{L}_{total} = \mathcal{L}_{pixel} + \mathcal{L}_{structure}. \tag{11}$$

C. Prior Condition Injection Module

After extracting the coarse-TCP $\mathbf{z}_c$ and the SMP $\mathbf{r}'_1$, we need to inject them into the U-net [52] via the prior condition injection module (PCIM), as shown in Fig. 4. To achieve this, we first downsample the input conditional features to match the multiscale representations of the network. Then,

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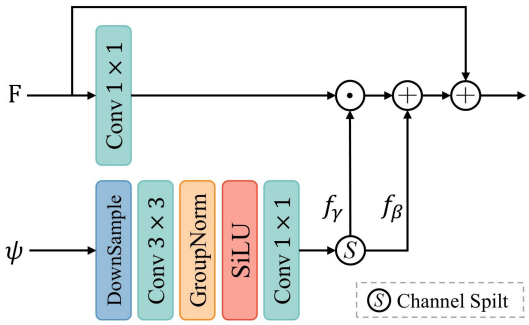


Fig. 4. Framework of PCIM. PCIM is designed to inject prior conditions into the U-net. The input condition ψ is first downsampled to match the feature $\mathbf{F}$. After doubling the channel dimension, the first half of the features f_γ is used as the scale, while the second half f_β serves as the shift to modulate the features.

these features undergo feature extraction and channel expansion before being split into two modulation tensors (f_γ, f_β). These learned modulation tensors are subsequently applied to adaptively perform an affine transformation on the spatial features, thereby injecting the conditional information into the feature maps. Specifically, the modulation tensors (f_γ, f_β) are derived from the prior conditions ψ through a mapping function $T : \psi \rightarrow (f_\gamma, f_\beta)$, formulated as

$$\begin{cases} (f_\gamma, f_\beta) = T(\psi) \\ \text{PCIM}(\mathbf{F} \mid f_\gamma, f_\beta) \quad = f_\gamma \odot \mathbf{F} + f_\beta \end{cases} \tag{12}$$

where $\mathbf{F}$ represents the input feature, and f_γ and f_β share the same dimensions as $\mathbf{F}$. $\odot$ denotes the elementwise multiplication, known as the Hadamard product. This approach enables the model to flexibly integrate prior conditions into the denoising process, allowing for more effective feature modulation and enhancing the overall generation quality.

The U-net architecture is composed of two key parts: an encoder and a decoder. Generally, the encoder should focus on extracting global style information while avoiding excessive encoding of spatial details. This ensures that the model prioritizes style-related feature extraction and minimizes interference from irrelevant spatial information [59]. To achieve this, we incorporate the PCIM into the encoder to inject low-frequency information $[\mathbf{C}_2, \mathbf{r}_1]$. Here, $\mathbf{C}_2$ provides the overall structure of the target image at a low spatial resolution, and $\mathbf{r}_1$ represents the mapping relationship between $\mathbf{C}$ and $\mathbf{F}$, effectively aiding the model in capturing global style information. In contrast, the decoder is responsible for faithfully reconstructing the spatial information extracted by the encoder to achieve high-quality image restoration. To further enhance feature representation and improve detail reconstruction, we introduce high-frequency details $[\mathbf{F}_1, \mathbf{z}_c]$ into the decoder using PCIM. Specifically, $\mathbf{F}_1$ provides rich texture details, while $\mathbf{z}_c$ captures the temporal change trends between reference and target time, assisting the model in recovering finer high-frequency details from the low-frequency information.

D. Difference-Aware Module

In this study, we leverage PENet to extract two priors while simultaneously generating a preliminary prediction $\mathbf{R}_{cnn}$,

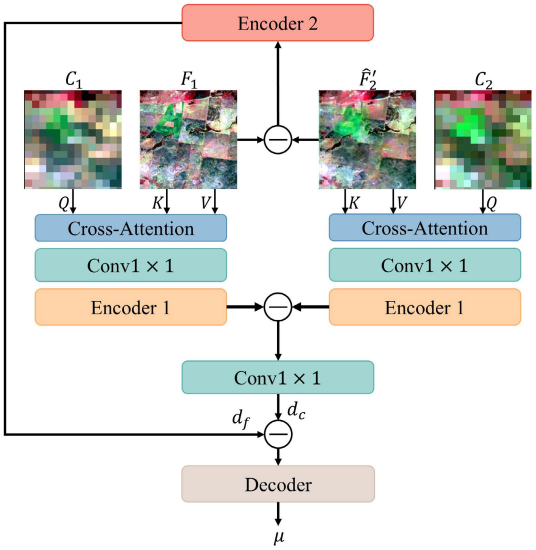


Fig. 5. Framework of DAM. DAM compensates for discrepancies in PENet output by leveraging temporal change consistency across resolutions. Given the preliminary reconstruction $\hat{\mathbf{F}}_2' = \mathbf{R}_{cnn} + \mathbf{C}_2$, the temporal change discrepancy between $(\mathbf{C}_1, \mathbf{C}_2)$ and $(\mathbf{F}_1, \hat{\mathbf{F}}_2')$ is estimated to refine the prediction. Cross-attention enhances coarse features to ensure more reliable representations. The final refinement term μ is derived by comparing the actual and predicted temporal change differences ($\mathbf{d}_c$ versus $\mathbf{d}_f$).

which can also serve as a guiding signal for the diffusion process. However, to prevent the model from over-relying on $\mathbf{R}_{cnn}$, we introduce the DAM that outputs a refinement term μ. The final guidance signal fed into the diffusion model is then formulated as $\mathbf{R}_{cnn} + \mu$. Since the TCP assumes that the change trend between coarse images is similar to that between fine images, we first derive the coarse-TCP from the known $\mathbf{C}_1$ and $\mathbf{C}_2$. Then, we obtain the fine-TCP using the predicted $\hat{\mathbf{F}}_2'$ and the known $\mathbf{F}_1$. By comparing the difference between these two priors, we estimate the prediction discrepancy.

As illustrated in Fig. 5, the module design follows a similar paradigm to PENet. Initially, we utilize $\mathbf{R}_{cnn}$ to compute an estimated reconstruction of $\mathbf{F}_2$ as $\hat{\mathbf{F}}_2' = \mathbf{R}_{cnn} + \mathbf{C}_2$. To further mitigate the resolution discrepancy between different scales, we incorporate cross-attention [26] to enhance the representations of both coarse-resolution inputs $(\mathbf{C}_1, \mathbf{C}_2)$ and their corresponding fine-resolution images $(\mathbf{F}_1, \hat{\mathbf{F}}_2')$, yielding refined feature representations $\bar{\mathbf{C}}_1$ and $\bar{\mathbf{C}}_2$

$$\begin{aligned} \bar{\mathbf{C}}_1 &= \text{Attention}(\mathbf{C}_1, \mathbf{F}_1, \mathbf{F}_1) + \mathbf{C}_1 \\ \bar{\mathbf{C}}_2 &= \text{Attention}(\mathbf{C}_2, \hat{\mathbf{F}}_2', \hat{\mathbf{F}}_2') + \mathbf{C}_2 \end{aligned} \tag{13}$$

where Attention is defined in (8). This way, it can ensure more reliable characteristics of $\mathbf{C_1}$ and $\mathbf{C_2}$ without introducing additional high-resolution information. Subsequently, we compute the actual coarse difference $\mathbf{d}_c$ by encoding the refined features using a weight-sharing encoder, denoted as Encoder_1

$$\mathbf{d}_c = \text{Encoder}_1(\bar{\mathbf{C}}_2) - \text{Encoder}_1(\bar{\mathbf{C}}_1). \tag{14}$$

Simultaneously, a separate encoder Encoder_2 is employed to compute the predicted difference $\mathbf{d}_f$

$$\mathbf{d}_f = \text{Encoder}_2(\hat{\mathbf{F}}_2' - \mathbf{F}_1). \tag{15}$$

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Finally, the residual difference $\mathbf{d}_f - \mathbf{d}_c$ is processed through a decoder to yield the discrepancy compensation term μ.

IV. Experiments

A. Data Description

In our experiments, we evaluate the proposed model on two classical public datasets, CIA and LGC [9]. To further assess model generalization and performance on more recent and diverse platforms, we also include the Sentinel-based WaterBank dataset [36] and the global E-SMILE dataset [12].

1) CIA Dataset: The CIA research area is located in the southern part of New South Wales, Australia, primarily focused on rice cultivation. The dataset includes 17 pairs of cloud-free MODIS-Landsat images from 2001 to 2002, with each pair measuring 6×2040×1720. The data collection period saw significant phenological changes, offering a valuable opportunity to evaluate performance in predicting temporal changes in vegetation phenology.

2) LGC Dataset: The LGC research area is located in the northern part of New South Wales, consisting of 14 pairs of cloud-free MODIS-Landsat images from 2004 to 2005, with each pair measuring 6 × 3200 × 3720. During the data collection period, a flood event caused significant changes in land cover, making the LGC dataset particularly suitable for testing performance in handling land-cover changes.

3) WaterBank Dataset: The WaterBank research area is located in northern Western Australia, characterized by a hot semiarid climate with distinct wet and dry seasons. The dataset includes 38 pairs of cloud-free Sentinel-2/Sentinel-3 Ocean and Land Colour Instrument (OLCI) images from January 9, 2019, to November 14, 2020, with each pair measuring 1800 × 1800. The area experiences frequent fire events and rapid vegetation changes throughout the year, providing a valuable opportunity to evaluate the performance of STF methods in capturing abrupt and heterogeneous temporal changes in vegetation patterns.

4) E-SMILE Dataset: The E-SMILE dataset is a large-scale, globally distributed dataset comprising MODIS-Landsat-8 image pairs collected across diverse geographic regions and all four seasons in 2020. The dataset contains 638 pairs for training and 450 pairs for testing, with each pair measuring 6 × 512 × 512. The test set includes challenges such as land-cover changes between phases and cloud contamination in the first phase, reflecting real-world data quality issues. With its wide coverage of seasonal dynamics, heterogeneous surface types, and land-cover variations, the E-SMILE dataset provides a valuable benchmark for evaluating the generalization ability and robustness of STF methods.

B. Experimental Settings

1) Dataset Setting: All datasets are partitioned into training and test subsets. For the CIA dataset, five image pairs acquired between March 17, 2002, and April 27, 2002, are used as the test set, while the remaining 12 pairs constitute the training set. For the LGC dataset, four image pairs collected from January 13, 2005, to March 2, 2005, serve as the test set, with the remaining ten pairs used for training. The reference image dates selected for testing are chosen from adjacent time periods to ensure fairness and reduce temporal bias. For the WaterBank dataset, five image pairs are randomly selected for testing, and the remaining 33 pairs are used for training. For the E-SMILE dataset, we follow the original setting by the authors, with 638 pairs used for training and 450 pairs for testing. A summary of the dataset configurations is provided in Table I.

TABLE I
Prediction and Reference Dates for Different Datasets

Dataset	Platform	Prediction Date	Reference Date
CIA [9]	Landsat 7-MODIS	Mar. 17th, 2002	Mar. 10th, 2002
		Apr. 2nd, 2002	Mar. 17th, 2002
		Apr. 11th, 2002	Apr. 2nd, 2002
		Apr. 18th, 2002	Apr. 11th, 2002
		Apr. 27th, 2002	Apr. 18th, 2002
LGC [9]	Landsat 5-MODIS	Jan. 13th, 2005	Jan. 29th, 2005
		Jan. 29th, 2005	Feb. 14th, 2005
		Feb. 14th, 2005	Mar. 2nd, 2005
		Mar. 2nd, 2005	Feb. 14th, 2005
WaterBank [36]	Sentinel 2-Sentinel 3	May 14th, 2019	May 4th, 2019
		Jul. 23rd, 2019	Jul. 3rd, 2019
		Apr. 28th, 2020	Mar. 24th, 2020
		May 3rd, 2020	Apr. 28th, 2020
		May 8th, 2020	May 3rd, 2020
E-SMILE [12]	Landsat 8-MODIS	/	/

2) Comparison Methods: To comprehensively assess the proposed model, we compare it with six representative methods, including one classical approach (FSDAF [44]), three generative models (GAN-STFM [20], MLFF-GAN [21], and STFDiff [12]), one frequency-domain-based model (ECPW-STFN [18]), and one spectral-optimization-oriented model (GSINet [19]).

For a fair comparison, all algorithms are evaluated under the same experimental settings. The hyperparameters for the classical methods are set according to their original articles, with the search similar pixel sliding window size fixed at 41 × 41 and the similarity threshold set to 20. For the deep learning-based methods, input images are cropped into 256 × 256 patches from the original images with a stride of 200. The diffusion time step T for STFDiff is set to 100.

3) Model Implementation: The proposed DPDiff is implemented using Pytorch 1.11.0 and Python 3.9.20, with the AdamW [60] optimizer, and trained on two NVIDIA GeForce RTX 3090 GPUs. During training, the date of the reference image is randomly selected from other dates within the training set. The input image size is set to 256 × 256, with a stride of 200. For PENet, the learning rate is set to 2×10^{-4}, with 300 training epochs and a batch size of 12. For the diffusion model, to validate the effectiveness of the proposed prior, we adopt the same U-net [52] architecture and noise schedule as RDDM [61], while integrating the PCIM before each up/downsampling operation. The number of encoder and decoder layers is set to 3, with a hidden dimension of 32. The learning rate is set to 8.8×10^{-4}, with a batch size of 4

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TABLE II

EXPERIMENTAL AVERAGE RESULTS ON TWO DATASETS. BOLD AND UNDERLINED TEXTS DENOTE THE BEST AND SECOND-BEST PERFORMANCES, RESPECTIVELY

Method	Parameters	CIA [9]				LGC [9]				WaterBank [36]				E-SMILE [12]			
		MAE ↓	RMSE ↓	SAM ↓	SSIM ↑	MAE ↓	RMSE ↓	SAM ↓	SSIM ↑	MAE ↓	RMSE ↓	SAM ↓	SSIM ↑	MAE ↓	RMSE ↓	SAM ↓	SSIM ↑
FSDAF [44]	-	0.0192	0.0296	0.1592	0.8464	0.0146	0.0206	0.1130	0.9128	0.0348	0.0387	0.2143	0.8756	0.0273	0.0385	0.1678	0.7690
GAN-STFM [20]	4.2×10^6	0.0190	0.0274	0.1489	0.8605	0.0139	0.0191	0.1101	0.9269	0.0076	0.0104	0.0679	0.9699	0.0354	0.0350	0.1408	0.8346
MLFF-GAN [21]	9.9×10^6	0.0179	0.0264	0.1463	0.8538	0.0125	0.0174	0.0992	0.9307	0.0207	0.0233	0.0955	0.9552	0.0418	0.0519	0.2036	0.7812
ECPW-STFN [18]	4.7×10^5	0.0190	0.0289	0.1550	0.8746	0.0217	0.0291	0.1681	0.9043	0.0188	0.0243	0.1310	0.9420	0.0418	0.0529	0.2098	0.7996
STFDiff [12]	7.5×10^6	0.0168	0.0267	0.1439	0.8686	0.0128	0.0179	0.1028	0.9255	0.0104	0.0136	0.0863	0.9648	0.0264	0.0351	0.1508	0.8298
GSINet [19]	3.0×10^6	0.0163	0.0233	0.1309	0.8762	0.0125	0.0175	0.1009	0.9355	0.0064	0.0091	0.0609	0.9730	0.0236	0.0314	0.1397	0.8378
DPDiff	5.3×10^6	**0.0159**	**0.0226**	**0.1268**	**0.8830**	**0.0121**	**0.0166**	**0.0930**	**0.9409**	**0.0064**	**0.0089**	**0.0582**	**0.9737**	**0.0211**	**0.0283**	**0.1311**	**0.8532**

and a total of 2000 training epochs. The diffusion process is conducted with 12 time steps and 5 sampling steps.

4) Evaluation Indicators: We employ four evaluation metrics to comprehensively assess method performance: mean absolute error (MAE), root-mean-square error (RMSE), spectral angle mapper (SAM) [62], and SSIM [63]. MAE and RMSE quantify overall spectral reconstruction accuracy, SAM evaluates spectral curve fidelity, and SSIM measures spatial structure preservation. Lower MAE, RMSE, and SAM values indicate better results, while a higher SSIM is preferred, with an ideal value of 1. For visualization, we present results using a false-color composite (near infrared response (NIR)–red–green).

C. Experimental Results

To ensure the fairness of comparison and the reproducibility of the experimental analysis, the average performance over all spectral channels is reported in the evaluation. Table II summarizes the overall results of different methods on the four datasets. It can be observed that diffusion-based models generally exhibit slower inference speed compared with several efficient counterparts. For instance, on the LGC dataset with a patch size of 340 × 400, DPDiff requires 31.47 s for inference—considerably faster than STFDiff (196 s) and comparable to ECPW-STFN (38.39 s)—yet still slower than the GAN-based approaches GAN-STFM (4.95 s) and MLFF-GAN (4.33 s), as well as the Transformer-based GSINet (10 s). This efficiency gap mainly stems from the iterative sampling mechanism of diffusion models, whose inference complexity can be approximated as $\mathcal{O}(T \cdot \mathcal{F})$, where T denotes the sampling steps and $\mathcal{F}$ is the cost of a single forward pass. In contrast, GAN- and Transformer-based STF methods typically adopt one-shot generation, resulting in lower latency. For large-scale remote sensing applications, the efficiency of DPDiff can be further improved via reduced-step sampling, sliding-window parallel inference, or lightweight backbone design to achieve a better efficiency–accuracy tradeoff. Nevertheless, regardless of the dataset used—whether the classical CIA and LGC datasets, the newly constructed WaterBank dataset derived from emerging satellite platforms, or the global-scale E-SMILE dataset—DPDiff consistently delivers the best fusion accuracy, demonstrating its robustness and generalization capability across diverse spatiotemporal scenarios. To comprehensively validate the effectiveness of the proposed model, we select the LGC and WaterBank datasets as representative benchmarks for in-depth comparative analysis.

1) Fusion Results on WaterBank: Table III presents the quantitative comparison of fusion results in the WaterBank area. The study area is mainly characterized by phenological changes, such as forest fires and variations in vegetation coverage. Due to the extremely large resolution gap (up to 30×) between the fine and coarse images, coupled with the high spatial heterogeneity caused by center-pivot irrigation, subtle changes cannot be directly captured from the coarse observations. This scenario poses a significant challenge for models in accurately predicting abrupt changes and highly heterogeneous regions. It can be observed that the traditional FSDAF method performs substantially worse than deep learning-based approaches under these conditions, further confirming that the resolution disparity has a pronounced impact on the final prediction accuracy. Since this region

TABLE III

QUANTITATIVE EVALUATION RESULTS OF STF METHODS IN THE WATERBANK AREA

Prediction Date	Model	MAE↓	RMSE↓	SAM↓	SSIM↑
May 14th, 2019	FSDAF	0.0155	0.0206	0.1334	0.9394
	GAN-STFM	0.0085	0.0115	0.0723	0.9719
	MLFF-GAN	0.0206	0.0246	0.1217	0.9532
	ECPW-STFN	0.0183	0.0227	0.1256	0.9514
	STFDiff	0.0110	0.0143	0.0858	0.9743
	GSINet	0.0070	0.0097	0.0645	0.9789
	DPDiff	**0.0062**	**0.0088**	**0.0551**	**0.9841**
Jul. 23rd, 2019	FSDAF	0.0368	0.0412	0.1117	0.9190
	GAN-STFM	0.0069	0.0095	0.0585	0.9684
	MLFF-GAN	0.0154	0.0178	0.0771	0.9630
	ECPW-STFN	0.0108	0.0146	0.0856	0.9611
	STFDiff	0.0101	0.0133	0.0754	0.9612
	GSINet	0.0060	0.0085	0.0501	0.9722
	DPDiff	**0.0059**	**0.0079**	**0.0469**	**0.9807**
Apr. 28th, 2020	FSDAF	0.0381	0.0413	0.2005	0.8661
	GAN-STFM	0.0088	0.0117	0.0758	0.9682
	MLFF-GAN	0.0286	0.0306	0.0959	0.9442
	ECPW-STFN	0.0213	0.0268	0.1421	0.9323
	STFDiff	0.0105	0.0140	0.0959	0.9604
	GSINet	**0.0071**	**0.0099**	**0.0676**	**0.9708**
	DPDiff	0.0075	0.0101	0.0694	0.9661
May 3rd, 2020	FSDAF	0.0673	0.0699	0.5092	0.7131
	GAN-STFM	0.0065	0.0093	0.0673	0.9706
	MLFF-GAN	0.0218	0.0245	0.0972	0.9543
	ECPW-STFN	0.0222	0.0291	0.1562	0.9349
	STFDiff	0.0096	0.0129	0.0909	0.9648
	GSINet	**0.0059**	**0.0086**	**0.0636**	**0.9712**
	DPDiff	0.0063	0.0093	0.0669	0.9647
May 8th, 2020	FSDAF	0.0163	0.0203	0.1167	0.9404
	GAN-STFM	0.0071	0.0099	0.0656	0.9706
	MLFF-GAN	0.0172	0.0193	0.0858	0.9613
	ECPW-STFN	0.0216	0.0281	0.1453	0.9304
	STFDiff	0.0106	0.0137	0.0837	0.9631
	GSINet	**0.0059**	0.0087	0.0586	0.9719
	DPDiff	0.0061	**0.0085**	**0.0530**	**0.9729**

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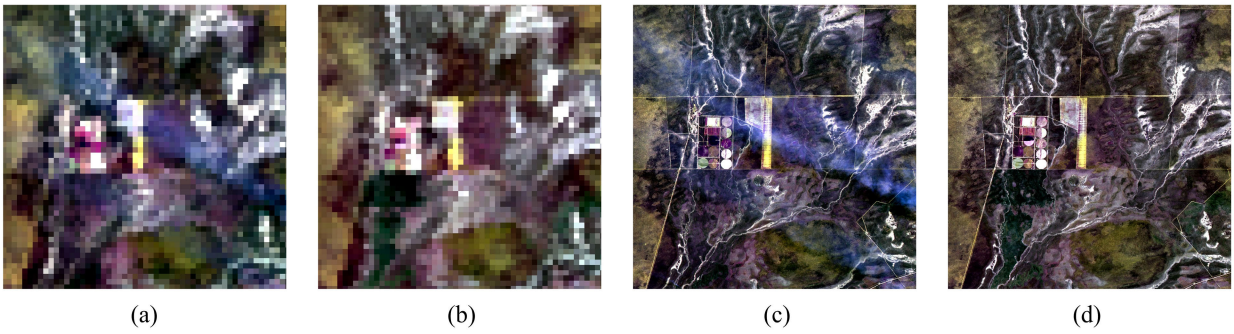


Fig. 6. Illustration of the WaterBank image pairs. (a) Coarse image on May 4, 2019. (b) Coarse image on May 14, 2019. (c) Fine image on May 4, 2019. (d) Fine image on May 14, 2019.

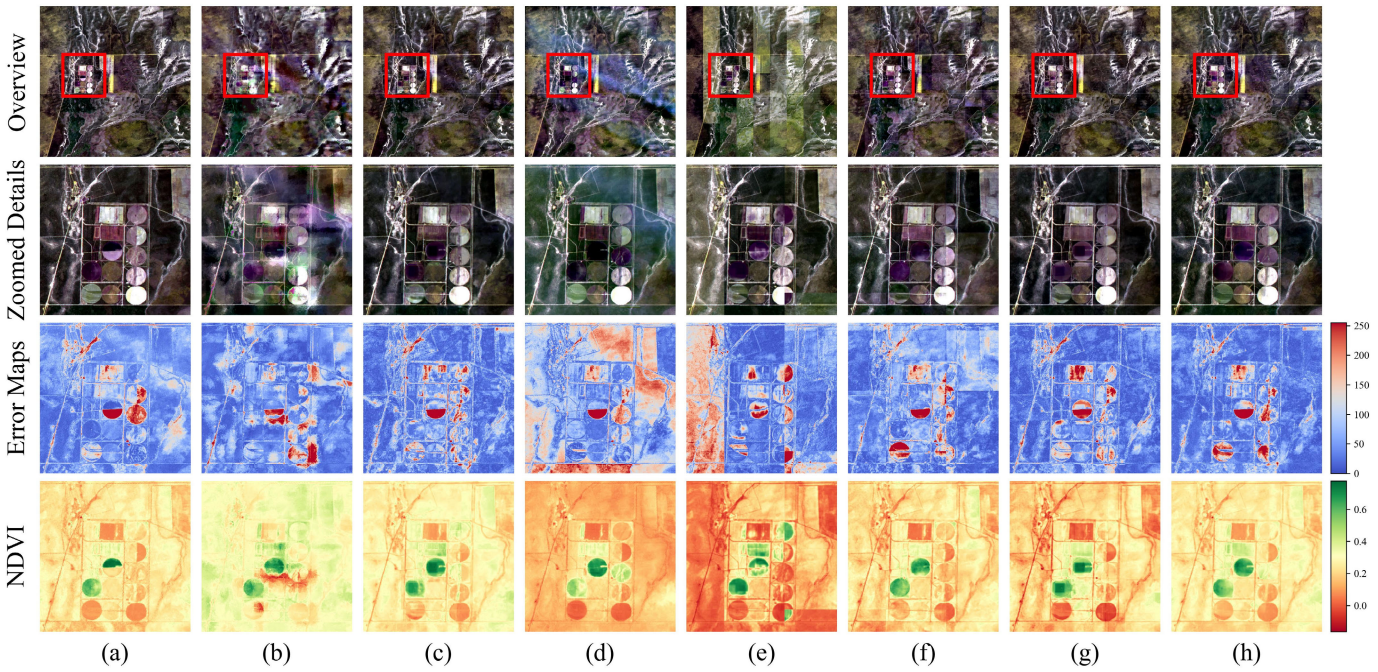


Fig. 7. Fusion results for the WaterBank area on May 14, 2019, using different STF methods. The last three rows correspond to the region in the first row marked with red rectangles, and the error map for GT is derived by computing the difference with the fine image at the reference time. (a) GT. (b) FSDAF. (c) GAN-STFM. (d) MLFF-GAN. (e) ECPW-STFN. (f) STFDiff. (g) GSINet. (h) DPDiff.

does not exhibit notable land-cover transitions, both GAN-STFM and MLFF-GAN achieve relatively high SSIM values. However, due to the frequent phenological variations, their performance in terms of MAE, RMSE, and SAM metrics is inferior to that of GSINet and STFDiff. The ECPW-STFN model shows large fluctuations across spectral bands, resulting in lower overall average accuracy. In contrast, our proposed method consistently achieves the best overall performance across all evaluation metrics, demonstrating that the introduced prior remains effective on the Sentinel 2–Sentinel 3 platform, particularly under large-resolution-scale conditions.

To facilitate a more intuitive comparison, we selected the predicted results on May 14 for detailed visualization. As illustrated in Fig. 6, the coarse and fine image pairs at the reference time (May 4) and the target time (May 14) are presented. The overall changes in this region are relatively minor, primarily manifested in the color variations of farmlands and center-pivot irrigation areas. The prediction results generated by different models are displayed in Fig. 7. The first row displays the predicted images, while the magnified view of the red rectangular region is presented in the second row. The third row visualizes the error maps, highlighting the differences between the predicted images and the GT, where redder areas indicate larger errors. Specifically, the GT error map is obtained by subtracting the fine image at the reference time from that at the target time, directly illustrating temporal changes. The fourth row presents NDVI maps [64], compared with the GT for a more intuitive assessment of the reconstruction performance of each model.

From the visual comparison, the traditional method produces relatively blurry predictions under large resolution disparities, whereas deep learning-based approaches demonstrate a stronger capability to adapt to such significant resolution differences. The main prediction challenge in the WaterBank area lies in the center-pivot irrigation regions, where it is difficult for models to capture such fine-grained

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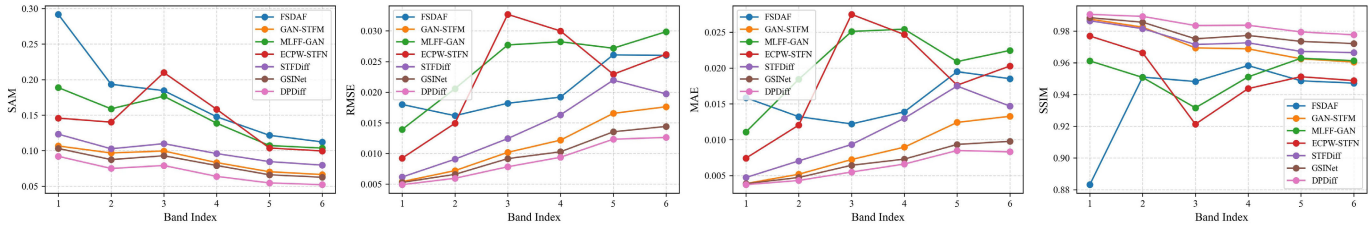


Fig. 8. Per-band results of different models on the WaterBank dataset. (a) SAM. (b) RMSE. (c) MAE. (d) SSIM. The proposed method achieves better accuracy and more stable performance across all bands.

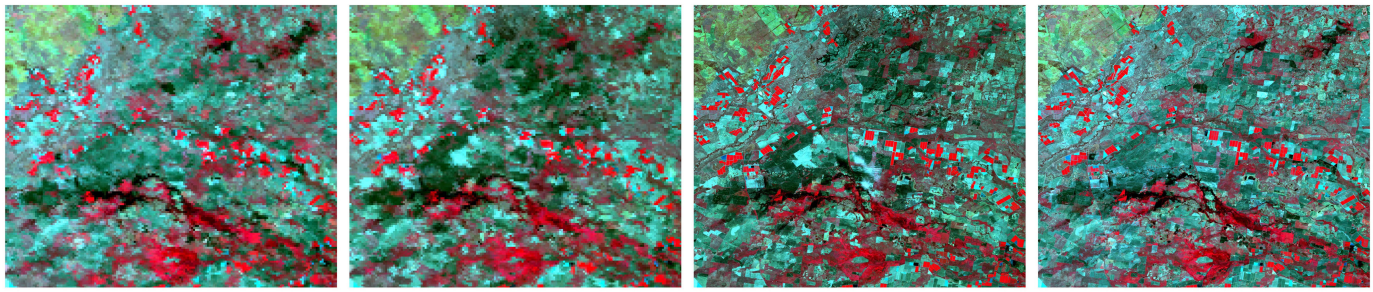


Fig. 9. Illustration of the LGC image pairs. (a) and (b) Coarse image on January 13, 2005. (c) and (d) Fine image on January 13, 2005.

changes from only two coarse inputs accurately. This issue can be clearly observed in the third row of the error maps, where all models exhibit noticeable prediction errors in these regions. Due to the stylized nature of the fine reference image, MLFF-GAN tends to generate predictions with a similar visual style (e.g., preserving smoke traces from fire-affected areas). The predictions of GAN-STFM and ECPW-STFM also resemble the reference fine image, which is particularly evident in the NDVI maps. Although STFDiff and GSINet produce overall accurate predictions, they still show limitations in restoring fine details compared with the proposed method, as also reflected in the NDVI maps shown in the fourth row. The proposed method achieves superior performance in both texture and structural restoration, while yielding the lowest reconstruction errors among all compared approaches. Furthermore, we visualize the per-band evaluation results of each method, as shown in Fig. 8. It is evident that the proposed method consistently achieves high prediction accuracy across all spectral bands while maintaining stable performance fluctuations, further validating the effectiveness and robustness of the proposed approach.

2) Fusion Results on LGC: Table IV presents the quantitative comparison results for the LGC area. The LGC region undergoes significant land-cover type transitions, making the prediction task more challenging than in the CIA and WaterBank areas. Besides, the 16× resolution difference poses a relatively smaller challenge compared with the WaterBank region, allowing the models to capture certain change priors and thus better exploit the generative capability of the diffusion model. For GAN-STFM and ECPW-STFN, the absence of the reference-time coarse image as input leads to degraded performance, particularly in regions undergoing significant surface changes, which is also reflected by a decline in quantitative accuracy metrics. MLFF-GAN, by modeling global relationships through AdaIN [65], maintains relatively stable performance. GSINet leverages the complementarity

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF STF METHODS IN THE LGC AREA (2005)

Prediction Date	Model	MAE↓	RMSE↓	SAM↓	SSIM↑
Jan. 13th	FSDAF	0.0174	0.0240	0.1278	0.8912
	GAN-STFM	0.0147	0.0199	0.1205	0.9154
	MLFF-GAN	0.0140	0.0187	0.1079	0.9172
	ECPW-STFN	0.0202	0.0270	0.1679	0.8983
	STFDiff	0.0145	0.0197	0.1136	0.9125
	GSINet	0.0132	0.0179	0.1085	0.9218
	DPDiff	**0.0122**	**0.0166**	**0.1022**	**0.9332**
Jan. 29th	FSDAF	0.0164	0.0232	0.1313	0.8981
	GAN-STFM	0.0165	0.0225	0.1253	0.9129
	MLFF-GAN	**0.0134**	**0.0192**	0.1129	0.9199
	ECPW-STFN	0.0244	0.0336	0.2005	0.8876
	STFDiff	0.0143	0.0204	0.1162	0.9128
	GSINet	0.0147	0.0207	0.1162	0.9195
	DPDiff	0.0161	0.0215	**0.1105**	**0.9226**
Feb. 14th	FSDAF	0.0125	0.0177	0.0983	0.9301
	GAN-STFM	0.0123	0.0172	0.0997	0.9377
	MLFF-GAN	0.0117	0.0161	0.0894	0.9425
	ECPW-STFN	0.0200	0.0274	0.1640	0.9128
	STFDiff	0.0112	0.0159	0.0915	0.9380
	GSINet	0.0115	0.0161	0.0931	0.9439
	DPDiff	**0.0104**	**0.0145**	**0.0828**	**0.9518**
Mar. 2nd	FSDAF	0.0122	0.0174	0.0946	0.9317
	GAN-STFM	0.0120	0.0168	0.0951	0.9415
	MLFF-GAN	0.0110	0.0155	0.0866	0.9431
	ECPW-STFN	0.0222	0.0286	0.1401	0.9185
	STFDiff	0.0112	0.0156	0.0874	0.9386
	GSINet	0.0107	0.0152	0.0857	0.9489
	DPDiff	**0.0097**	**0.0138**	**0.0764**	**0.9561**

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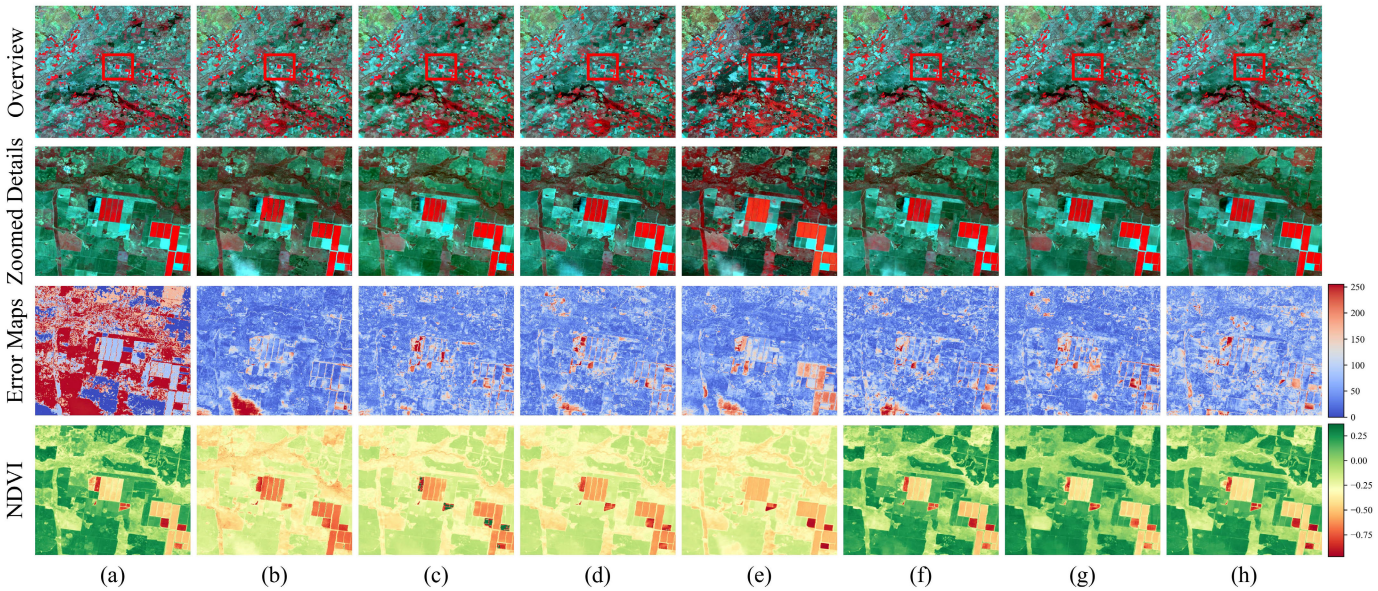


Fig. 10. Fusion results for the LGC area on January 13, 2005, using different STF methods. The last three rows correspond to the region in the first row marked with red rectangles, and the error map for GT is derived by computing the difference with the fine image at the reference time. (a) GT. (b) FSDAF. (c) GAN-STFM. (d) MLFF-GAN. (e) ECPW-STFN. (f) STFDiff. (g) GSINet. (h) DPDiff.

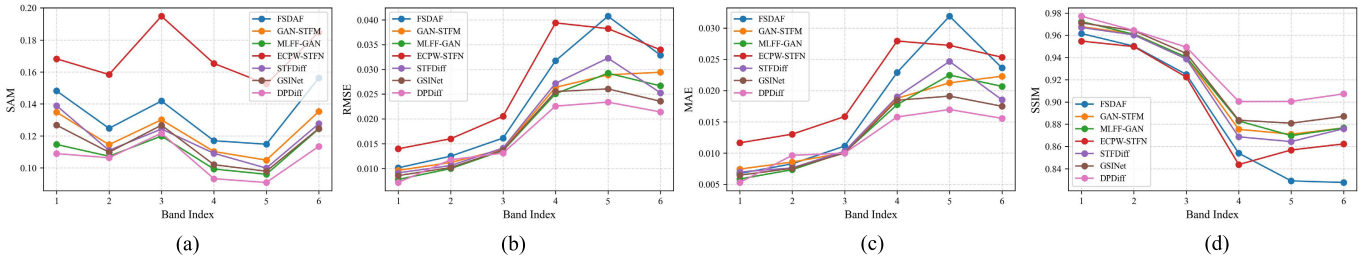


Fig. 11. Per-band results of different models on the LGC dataset. (a) SAM. (b) RMSE. (c) MAE. (d) SSIM. The proposed method achieves better accuracy and more stable performance across all bands.

among spectral bands to enhance prediction accuracy to some extent; however, its improvement remains limited in areas with drastic temporal variations. STFDiff directly employs the known conditions as modulation inputs but fails to effectively exploit the underlying dependencies among inputs, leading to suboptimal results in complex spatiotemporal scenarios. In contrast, the proposed DPDiff consistently achieves superior fusion performance under these challenging conditions, demonstrating stronger robustness and adaptability to dynamic surface changes.

For a more intuitive comparison, we select the prediction results of January 13 to showcase fusion details. Fig. 9 presents the coarse and fine image pairs corresponding to the reference time of January 29 and the target time of January 13, showing significant overall changes. Fig. 10 displays the predicted results from different models. Traditional methods exhibit large errors in change areas and weak spectral restoration accuracy. The fusion-based ECPW-STFN method produces a prediction that is closer to the reference image, which leads to higher prediction errors. Compared with the WaterBank dataset, the LGC region exhibits more significant variations in land-cover types. Consequently, the absence of change information leads to incomplete structural details in the predicted images generated by MLFF-GAN and GAN-STFM. Similarly, GSINet and STFDiff suffer from certain distortions along the boundaries of cropland areas. Furthermore, as illustrated in Fig. 11, we visualize the reconstruction performance of each model across different spectral bands to assess their spectral prediction accuracy. It can be observed that the differences among all methods are relatively small in the first three bands (i.e., the RGB channels), while the discrepancies become more pronounced in the last three bands (NIR, MIR1, and MIR2). This is primarily attributed to the inherent differences in spatial resolution and spectral characteristics among these bands. Although GSINet addresses this issue through grouped feature fusion, our proposed method still outperforms GSINet, demonstrating the potential of diffusion-based frameworks and the introduced priors in multisource remote sensing image fusion tasks.

D. Uncertainty Analysis

In practical applications, temporal uncertainty implies that the reference images are not always temporally close to the target date, which poses challenges to model stability and prediction accuracy. To quantitatively analyze the influence

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TABLE V

QUANTITATIVE EVALUATION RESULTS FOR THE WATERBANK AREA ON MAY 14, 2019, USING DIFFERENT REFERENCE IMAGES

Model	Reference Date	MAE↓	RMSE↓	SAM↓	SSIM↑
FSDAF	Jan. 9th	0.1221	0.1246	1.1165	0.0492
	Mar. 10th	0.0677	0.0702	0.2985	0.6558
	May. 4th	0.0155	0.0206	0.1334	0.9394
	Jun. 23rd	0.0172	0.0229	0.1554	0.9388
GAN-STFM	Jan. 9th	0.0097	0.0135	0.0875	0.9554
	Mar. 10th	0.0084	0.0117	0.0759	0.9667
	May. 4th	0.0083	0.0112	0.0718	0.9719
	Jun. 23rd	0.0066	0.0093	0.0603	0.9766
MLFF-GAN	Jan. 9th	0.0273	0.0296	0.1245	0.9420
	Mar. 10th	0.0328	0.0357	0.1485	0.9283
	May. 4th	0.0291	0.0316	0.1306	0.9416
	Jun. 23rd	0.0235	0.0277	0.1485	0.9333
ECPW-STFN	Jan. 9th	0.0123	0.0167	0.1038	0.9626
	Mar. 10th	0.0236	0.0292	0.1394	0.9360
	May. 4th	0.0230	0.0275	0.1363	0.9433
	Jun. 23rd	0.0149	0.0188	0.1094	0.9650
STFDiff	Jan. 9th	0.0135	0.0179	0.1129	0.9503
	Mar. 10th	0.0118	0.0156	0.0981	0.9637
	May. 4th	0.0110	0.0143	0.0908	0.9746
	Jun. 23rd	0.0103	0.0135	0.0875	0.9747
GSINet	Jan. 9th	0.0092	0.0131	0.0913	0.9577
	Mar. 10th	0.0076	0.0109	0.0755	0.9715
	May. 4th	0.0069	0.0099	0.0672	0.9787
	Jun. 23rd	0.0061	0.0086	0.0587	0.9818
DPDiff	Jan. 9th	0.0078	0.0111	0.0731	0.9672
	Mar. 10th	0.0067	0.0096	0.0628	0.9758
	May. 4th	0.0062	0.0088	0.0511	0.9841
	Jun. 23rd	0.0055	0.0078	0.0488	0.9838

TABLE VI

QUANTITATIVE EVALUATION RESULTS FOR THE LGC AREA ON JANUARY 13, 2005, USING DIFFERENT REFERENCE IMAGES

Model	Reference Date	MAE↓	RMSE↓	SAM↓	SSIM↑
FSDAF	Dec. 12th	0.0247	0.0326	0.1952	0.8097
	Dec. 28th	0.0157	0.0218	0.1364	0.8880
	Jan. 29th	0.0174	0.0240	0.1278	0.8912
	Mar. 2nd	0.0163	0.0225	0.1316	0.8884
GAN-STFM	Dec. 12th	0.0173	0.0237	0.1497	0.8727
	Dec. 28th	0.0145	0.0198	0.1222	0.9130
	Jan. 29th	0.0142	0.0192	0.1174	0.9160
	Mar. 2nd	0.0147	0.0202	0.1247	0.9148
MLFF-GAN	Dec. 12th	0.0172	0.0234	0.1450	0.8590
	Dec. 28th	0.0147	0.0197	0.1158	0.9075
	Jan. 29th	0.0142	0.0190	0.1111	0.9151
	Mar. 2nd	0.0141	0.0195	0.1167	0.9159
ECPW-STFN	Dec. 12th	0.0348	0.0439	0.2370	0.8314
	Dec. 28th	0.0237	0.0310	0.1804	0.8897
	Jan. 29th	0.0202	0.0270	0.1679	0.8983
	Mar. 2nd	0.0243	0.0333	0.2107	0.8829
STFDiff	Dec. 12th	0.0146	0.0202	0.1285	0.8862
	Dec. 28th	0.0155	0.0206	0.1163	0.9128
	Jan. 29th	0.0145	0.0198	0.1138	0.9124
	Mar. 2nd	0.0153	0.0212	0.1279	0.9039
GSINet	Dec. 12th	0.0165	0.0229	0.1456	0.8773
	Dec. 28th	0.0133	0.0183	0.1146	0.9193
	Jan. 29th	0.0131	0.0178	0.1080	0.9221
	Mar. 2nd	0.0134	0.0186	0.1154	0.9209
DPDiff	Dec. 12th	0.0152	0.0208	0.1318	0.8953
	Dec. 28th	0.0123	0.0169	0.1044	0.9326
	Jan. 29th	0.0122	0.0166	0.1022	0.9332
	Mar. 2nd	0.0128	0.0175	0.1091	0.9284

of temporal intervals between reference and target dates, and to verify the effectiveness of the proposed temporal and spatial priors, we perform an uncertainty analysis across varying reference-time configurations. In general, inferring past states from current observations is relatively straightforward. However, in practical STF applications, the more realistic and challenging task is to predict the current or future state based on historical observations. Therefore, this study mainly focuses on evaluating model performance when using pretarget images as References, while several posttarget cases are also included to assess the model's capability of generalization. Specifically, experiments are conducted on the WaterBank and LGC datasets. For the WaterBank dataset, the target date is set to May 14, 2019. Three historical reference dates (January 9, March 10, and May 4) and one future reference date (June 23) are selected for evaluation. For the LGC dataset, considering its limited temporal coverage, two pretarget reference images (December 12 and 28, 2004) and two posttarget images (January 29 and March 2, 2005) are used. Notably, the reference image acquired on December 12, 2004, records a sudden flooding event in the LGC region, providing a valuable case for evaluating the ability of each method to handle abrupt environmental changes. The quantitative results are presented in Tables V and VI.

Specifically, all methods exhibit certain fluctuations across different reference times, which are primarily reflected in the SSIM metric. Among them, the proposed DPDiff consistently outperforms all other approaches under all temporal configurations, demonstrating superior stability and accuracy. However, the traditional method FSDAF suffers from significant performance degradation as the temporal interval between reference and target dates increases, particularly in the WaterBank region, where the resolution discrepancy is pronounced. Therefore, the following analysis focuses on deep learning-based methods. For clarity, the corresponding boxplots are shown in Fig. 12, where different colors indicate the performance of the six fusion models. The top and bottom of each box represent the maximum and minimum test results for different reference times, while the horizontal line within the box indicates the median of each group. Overall, DPDiff demonstrates consistently high prediction accuracy across different reference times, whereas other methods show greater variability.

For the WaterBank region, where both the spatial resolution discrepancy and the temporal interval are large, the MLFF-GAN and ECPW-STFN methods, which heavily rely on fine reference images, experience a marked decline in performance, and exhibit considerable fluctuations across different reference dates. A similar trend can also be observed for GSINet. Although GAN-STFM maintains relatively stable results, its overall performance remains inferior to that of the diffusion-based approaches. Meanwhile, STFDiff shows notable variations in the SSIM metric, suggesting that the absence of explicit SMPs leads to significant performance

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TABLE VII
ABLATION RESULTS OF DPDIFF

ID	DPDiff				LGC Dataset				WaterBank Dataset			
	DAM	PCIM	TCP	SMP	MAE↓	RMSE↓	SAM↓	SSIM↑	MAE↓	RMSE↓	SAM↓	SSIM↑
1		✓	✓	✓	0.0126	0.0189	0.0959	0.9399	0.0074	0.0099	0.0648	0.9715
2	✓		✓	✓	0.0122	0.0170	0.0967	0.9383	0.0078	0.0103	0.0664	0.9719
3					0.0126	0.0177	0.1006	0.9341	0.0072	0.0099	0.0642	0.9692
4	✓	✓	✓		0.0129	0.0179	0.1005	0.9343	0.0073	0.0099	0.0654	0.9707
5	✓	✓		✓	0.0122	0.0169	0.0967	0.9386	0.0070	0.0095	0.0652	0.9717
6	✓	✓	✓	✓	**0.0121**	**0.0166**	**0.0930**	**0.9409**	**0.0064**	**0.0089**	**0.0582**	**0.9737**

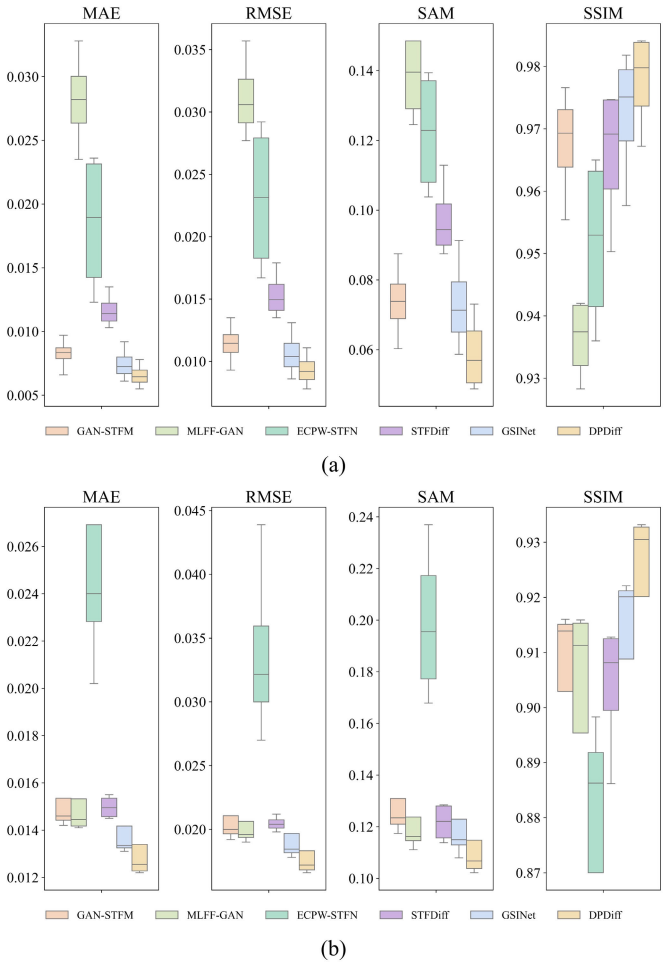


Fig. 12. Quantitative evaluation of different models with varying reference images using box plots. (a) WaterBank area on May 14, 2019. (b) LGC area on January 13, 2005.

degradation of conventional CNN-based models under temporal uncertainty. In contrast, the proposed DPDiff achieves the highest prediction accuracy while maintaining low variability, demonstrating superior robustness to both spatial and temporal discrepancies.

For the LGC dataset, where temporal changes are more frequent and complex, the performance differences among models become more pronounced. Notably, ECPW-STFN exhibits the largest fluctuation, indicating that the lack of TCPs causes traditional CNN models to degrade sharply under uncertain temporal conditions. Although STFDiff achieves the smallest fluctuation, it still produces lower accuracy, highlighting the importance of fully exploiting the relationships among input images to guide fusion. By contrast, DPDiff consistently maintains state-of-the-art performance across all reference times. Overall, DPDiff not only excels in spectral recovery and structural reconstruction but also effectively mitigates the strict temporal dependence that constrains conventional and most existing deep learning-based methods.

E. Ablation Study

To validate the effectiveness of the proposed components—including the SMP, TCP, DAM, and PCIM—a series of ablation experiments was conducted on the LGC and WaterBank datasets, as shown in Table VII. Experiments 1–5 were designed to isolate the contributions of DAM, PCIM, SMP, and TCP, as well as to evaluate the impact of replacing modulated features with the fine image $\mathbf{F}_1$ and coarse image $\mathbf{C}_2$. Experiment 6 integrates all components to demonstrate the full capability of the proposed framework.

The results reveal that the DAM module improves MAE and SAM by 4% and 3%, respectively, on the LGC dataset, and by 14% and 10% on the WaterBank dataset, highlighting its role in enhancing detail reconstruction under large resolution gaps. Incorporating PCIM further boosts all four evaluation metrics across both datasets, confirming its importance in effectively injecting prior information. Even without any priors or auxiliary modules, the diffusion model maintains competitive performance compared with mainstream methods, underscoring its inherent potential for STF. Nevertheless, removing these modules results in performance drops of 8% in MAE, 8.8% in RMSE, 9% in SAM, and 0.55% in SSIM, indicating that hard conditioning can hinder fine-detail generation—especially under frequent land-cover changes, as observed in LGC. The SMP prior enhances SSIM by 0.7% and 0.3% on the LGC and WaterBank datasets, respectively, while the TCP prior contributes improvements of 4% and 11% in SAM, further validating the complementary roles of spatial and temporal priors.

To visually demonstrate the contributions of each component in the model, we present the fusion results for the LGC area on January 13, 2005, as shown in Fig. 13. The figure consists of four rows, each highlighting different aspects of the results. The first row presents the reference (GT) image

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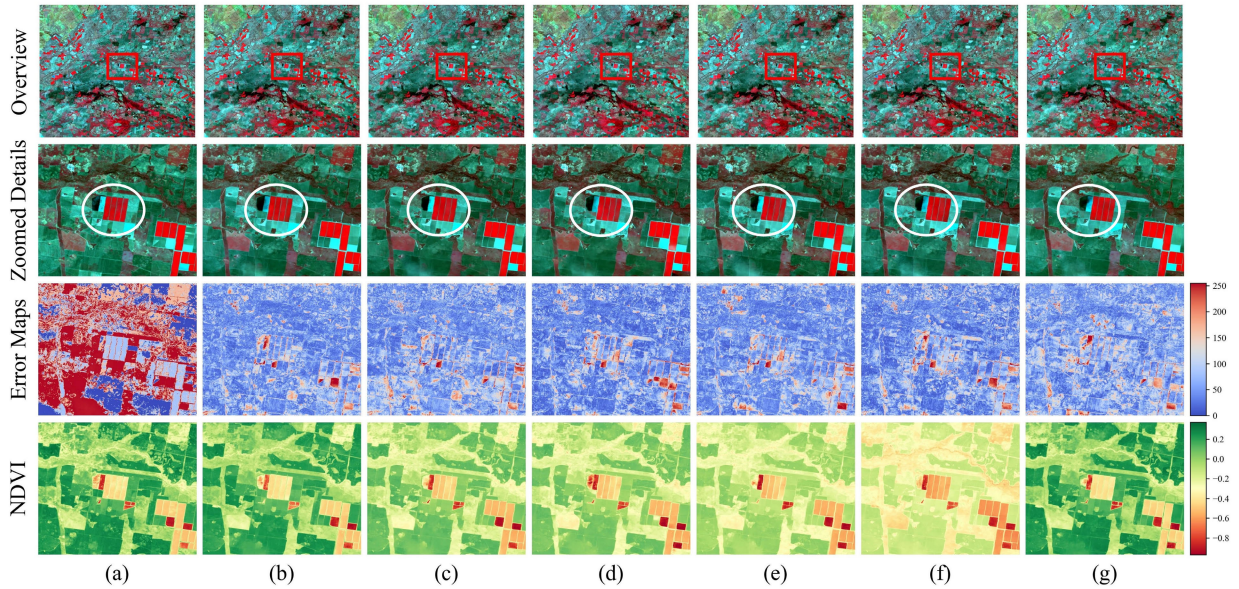


Fig. 13. Ablation study results for the LGC area on January 13, 2005. The last three rows correspond to the region in the first row marked with red rectangles, and the error map for GT is derived by computing the difference with the fine image at the reference time. (a) GT. (b) Ablation 1. (c) Ablation 2. (d) Ablation 3. (e) Ablation 4. (f) Ablation 5. (g) DPDiff.

for comparison. In the second row of zoomed-in view detail, the generated farmland edges—as highlighted by the white circles—show evident distortions when prior constraints are not applied [Fig. 13(b)–(e)], whereas the results with the SMP [Fig. 13(f) and (g)] retain clearer and more consistent structures. The third row presents the error maps, which reveal that directly enforcing known images as constraints does not necessarily reduce error; instead, it introduces additional distortions, particularly along object boundaries [Fig. 13(b) and (d)]. This supports our earlier analysis that excessive reliance on explicit constraints can hinder the generation of fine details. In contrast, the inclusion of DAM further decreases the reconstruction error [Fig. 13(b) and (g)]. Finally, the fourth row presents the NDVI maps. A direct comparison between Fig. 13(e) and (f) reveals that, in the LGC region, the absence of TCP guidance substantially compromises spectral reconstruction accuracy, whereas the lack of SMP predominantly deteriorates the precision of edge prediction.

In conclusion, the ablation study confirms that incorporating the spatiotemporal prior constraints in the diffusion process significantly improves model performance, especially in terms of fine-detail preservation and texture reconstruction. The introduction of priors helps maintain structural integrity, as observed in the farmland edge details and NDVI distributions. Additionally, the DAM effectively compensates for errors introduced by the PENet, leading to an overall performance boost in both quantitative metrics and visual quality.

V. Conclusion

This article has introduced DPDiff, a spatiotemporal DPDiff framework for STF. Unlike previous methods that rely solely on known images, DPDiff leverages a lightweight PENet to derive spatiotemporal dual priors, allowing for more precise control over the diffusion process. By incorporating residual diffusion, the method models the coarse image as a degraded version of the fine image, improving reconstruction accuracy. This approach enables DPDiff to operate robustly across varying temporal spans without the strict temporal alignment required by conventional methods, making it well-suited to handle spatiotemporal uncertainty. Furthermore, the extracted priors can be easily integrated into other diffusion frameworks, highlighting the adaptability and broad application potential. The effectiveness of DPDiff has been validated on two historical datasets, one contemporary dataset, and one global dataset, demonstrating its superior performance in STF tasks.

With ongoing advancements in diffusion-based architectures, computational efficiency remains a significant challenge. Although DPDiff delivers high-quality results with only five sampling steps, its inference speed still falls short of that achieved by conventional deep learning methods. Future work will focus on optimizing both the PENet and the U-net architecture to further enhance computational efficiency and improve adaptability for STF tasks. Additionally, further research is needed to boost the generalization ability of the model across diverse datasets and to handle unseen temporal gaps more effectively.

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